QIC 890: Quantum Complexity Theory
Problem Set 2

Due Friday, November 13, 2026

Problem 1

A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is contained in the complexity class NQP if there exists a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits (over our standard gate set $\{H, S, \text{TOF}\}$), where each Q_x has no input qubits and one output qubit, such that these implications are true.

$$x \in A_{\text{yes}} \implies \Pr(Q_x \text{ accepts}) > 0$$
$$x \in A_{\text{no}} \implies \Pr(Q_x \text{ accepts}) = 0$$

Prove that a given promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is in NQP if and only if there exists a GapP function g such that

$$x \in A_{\text{yes}} \implies g(x) \neq 0$$
$$x \in A_{\text{no}} \implies g(x) = 0$$

for all $x \in \Sigma^*$.

Problem 2

Consider the following restricted version of the local Hamiltonian problem, where the Hamiltonian is a diagonal matrix.

k-LOCAL DIAGONAL HAMILTONIAN (k-LDH)

Input Diagonal Pauli operations $P_1, \ldots, P_m$ on n qubits, each having weight at most k and each expressed as a string over the alphabet $\{\mathbb{1}, Z\}$; rational number coefficients $\gamma_1, \ldots, \gamma_m \in [-1, 1]$; and a positive integer K written in unary.

Yes $\lambda_{\min}(\gamma_1 P_1 + \cdots + \gamma_m P_m) \leq -1/K$.

No $\lambda_{\min}(\gamma_1 P_1 + \cdots + \gamma_m P_m) \geq 1/K$.

Prove that 3-LDH is NP-complete.

Problem 3

Let A and B be languages over an alphabet Σ, which we can identify with promise problems $(A, \Sigma^* \setminus A)$ and $(B, \Sigma^* \setminus B)$ as usual. Also let p be a polynomial resource bound, and assume that $A, B \in \mathrm{QMA}_p(2/3, 1/3)$.

(a) Prove that $A \cap B \in \mathrm{QMA}_{2p}(2/3, 1/3)$.

(b) Prove that $A \cup B \in \mathrm{QMA}_p(2/3, 1/3)$.

Problem 4

Consider a complexity class PostBQP/qpoly, in which a polynomial-length quantum advice state is provided to a quantum computation just like in BQP/qpoly, but this time the quantum computation can use postselection.

(a) Formulate a precise definition of PostBQP/qpoly by adapting the definitions of BQP/qpoly (Definition 11.5) and PostBQP (Definition 7.1) in a natural way.

(b) Prove that PostBQP/qpoly includes every language.